Technology Stack

Date	28 JUNE 2026
Team ID	xxxxxx
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML5, CSS3, Bootstrap, JavaScript*	*Used to build a responsive, user-friendly web interface for farmers to enter soil and environmental parameters and view crop recommendations.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python, Flask*	*Flask handles HTTP requests, connects the frontend with the machine learning model, processes user input, and returns crop prediction results efficiently*.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*CSV Dataset (Crop Recommendation Dataset)*	*Stores historical agricultural data containing soil nutrients, weather conditions, and crop labels used for training and prediction.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Local Flask Server / Render / PythonAnywhere*	*Provides an environment to deploy and access the web application with minimal configuration and easy scalability.*
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	*Git & GitHub*	*Used to maintain source code, track project changes, collaborate among team members, and manage different project versions.*

6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Scikit-learn, Pandas, NumPy, Matplotlib, Anaconda, Jupyter Notebook, VS Code*	*Scikit-learn is used for machine learning model development, Pandas and NumPy for data preprocessing, Matplotlib for visualization, Jupyter Notebook for model training, Anaconda for package management, and VS Code for application development.*